# MILITARY SPECIFICATION SHEET

## INSULATION, PLASTIC, LAMINATED, THERMOSETTING, GLASS-CLOTH, PHENOLIC-RESIN (GPG)

*This specification is approved for use by all departments and agencies of the Department of Defense.*

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

**REQUIREMENTS:**

**Construction:** The material shall consist of a woven glass cloth properly impregnated and bonded with phenolic-resin matrix and processed to conform to this specification.

**Intended use:** The laminated material covered by this specification is a material with high impact and flexural strength. It has good electrical properties under dry conditions. It is used for motor insulation and in other applications where high strength and good electrical properties are required, even at fairly high temperatures.

**Forms:** Sheets, rolled tubes, and rods are recommended forms of laminated material covered by this specification.

**Temperature indices:** The laminated material covered by this specification has a mechanical strength test index of 170, and an electrical strength test index of 140.

**Dimensions and part numbers:**

*Thickness of sheets:* The thickness of sheets, permissible variations, and the applicable part number shall be as specified in table I.

AMSC N/A FSC 5970
**DISTRIBUTION STATEMENT A.** Approved for public release; distribution is unlimited.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:00Z
Check the source to verify that this is the current version before use.

TABLE I. *Thickness of laminated sheets.*

| Part number | Nominal thickness (inches) | Permissible variations (± inch) |
|---|---|---|
| M24768/18-S-1 | 0.010 | 0.002 |
| M24768/18-S-2 | 0.015 | 0.003 |
| M24768/18-S-3 | 0.020 | 0.004 |
| M24768/18-S-4 | 0.025 | 0.005 |
| M24768/18-S-5 | 0.031 | 0.065 |
| M24768/18-S-6 | 0.047 | 0.0075 |
| M24768/18-S-7 | 0.062 | 0.0075 |
| M24768/18-S-8 | 0.094 | 0.009 |
| M24768/18-S-9 | 0.125 | 0.012 |
| M24768/18-S-10 | 0.156 | 0.015 |
| M24768/18-S-11 | 0.188 | 0.019 |
| M24768/18-S-12 | 0.219 | 0.021 |
| M24768/18-S-13 | 0.250 | 0.022 |
| M24768/18-S-14 | 0.312 | 0.026 |
| M24768/18-S-15 | 0.375 | 0.030 |
| M24768/18-S-16 | 0.438 | 0.033 |
| M24768/18-S-17 | 0.500 | 0.036 |
| M24768/18-S-18 | 0.625 | 0.040 |
| M24768/18-S-19 | 0.750 | 0.043 |
| M24768/18-S-20 | 0.875 | 0.046 |
| M24768/18-S-21 | 1.000 | 0.049 |
| M24768/18-S-22 | 1.125 | 0.053 |
| M24768/18-S-23 | 1.250 | 0.055 |
| M24768/18-S-24 | 1.375 | 0.058 |
| M24768/18-S-25 | 1.500 | 0.061 |
| M24768/18-S-26 | 1.625 | 0.064 |
| M24768/18-S-27 | 1.750 | 0.067 |
| M24768/18-S-28 | 1.875 | 0.070 |
| M24768/18-S-29 | 2.000 | 0.073 |
| M24768/18-S-30 | 2.125 | 0.079 |
| M24768/18-S-31 | 2.500 | 0.085 |
| M24768/18-S-32 | 2.750 | 0.091 |
| M24768/18-S-33 | 3.000 | 0.097 |
| M24768/18-S-34 | 3.500 | 0.110 |
| M24768/18-S-35 | 4.000 | 0.122 |
| M24768/18-S-36 | 4.500 | 0.134 |
| M24768/18-S-37 | 5.000 | 0.142 |
| M24768/18-S-38 | 5.500 | 0.156 |
| M24768/18-S-39 | 6.000 | 0.172 |
| M24768/18-S-40 | 6.500 | 0.180 |
| M24768/18-S-41 | 7.000 | 0.195 |
| M24768/18-S-42 | 7.500 | 0.210 |
| M24768/18-S-43 | 8.000 | 0.216 |
| M24768/18-S-44 | 8.500 | 0.232 |
| M24768/18-S-45 | 9.000 | 0.240 |
| M24768/18-S-46 | 9.500 | 0.255 |
| M24768/18-S-47 | 10.000 | 0.270 |
| M24768/18-S-48 | —[1] | —[2] |

See footnotes on next page.



Source: https://assist.dla.mil -- Downloaded: 202610-20T02:00Z
Check the source to verify that this is the current version before use.

[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/18-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

*Diameter of rolled round tubes:* The range of sizes for rolled round tubes shall be as specified in table II. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a rolled round tube with an inside diameter of 0.125 inch and outside diameter of 0.156 inch is: M24768/18-TRR-0.125/0.156. The wall thickness tolerances for rolled round tubes with an inside diameter up to 4 inches shall be as specified in table III.

TABLE II. *Range of sizes for rolled round tubes.*

| Inside diameter (inches) | | Outside diameter (inches) | | Wall thickness (inch) | | Maximum ratio of wall thickness to inside diameter |
|---|---|---|---|---|---|---|
| Minimum | Maximum | Minimum | Maximum | Minimum | Maximum | |
| 0.250 | 48.000 | 0.281 | 50.000 | 0.016 | 1.000 | 1/2 |

TABLE III. *Wall thickness tolerances.*

| Wall thickness (inch) | Tolerances (± inch) |
|---|---|
| 0.010 to 0.015 | – |
| 0.016 to 0.030 | 0.006 |
| 0.031 to 0.061 | 0.008 |
| 0.062 to 0.124 | 0.009 |
| 0.125 to 0.249 | 0.011 |
| 0.250 to 0.500 | 0.013 |

*Diameter of rods:* The range of sizes and tolerances for rods shall be as specified in table IV. The outside diameter of the rod shall be included in the part number. An example of a part number for a rod with an outside diameter of 0.250 inch is: M24768/18-R-0.250.

TABLE IV. *Size and tolerances for rods.*

| Diameter (inches) | Tolerances (± inch) |
|---|---|
| 0.250 to 1.999 | 0.005 |
| 2.000 | 0.008 |



Source: https://assist.dla.mil -- Downloaded: 20261O-20T02:00Z
Check the source to verify that this is the current version before use.

**Performance requirements:** Performance requirements for laminated materials shall be as specified in tables V, VI, and VII.

The following performance requirements do not apply to this plastic laminate:

a. Dielectric breakdown voltage
b. Dielectric strength
c. Surface resistance
d. Volume resistivity
e. Silicone content
f. Arc resistance
g. Thermal endurance
h. Flammability
i. Tracking resistance
j. Dielectric constant
k. Dissipation factor.

**Testing:** The laminated material covered by this specification shall be tested under first article inspection requirements.



Source: https://assist.dla.mil -- Downloaded: 202610-20T02:00Z
Check the source to verify that this is the currentversion before use.

TABLE V. *Performance requirements for sheets.*

| Requirement | Cond. | Test para. | Unit | Thickness (inches)[1] | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 0.031 | 0.062 | 0.094 | 0.125 | 0.188 | 0.250 | 0.500 | 0.750 | 1.000 | 1.001 – Max. |
| Dielectric breakdown voltage | A<br>D-48/50<br>D-336/50 | 4.6.4 | Min. kV | | | | | | | | | | |
| Dielectric strength | A<br>D-48/50 | 4.6.5 | Min. volts per mil | | | | | | | | | | |
| Dielectric constant | A<br>D-24/23,<br>D-48/50 | 4.6.5 | Max. | | | | | | | | | | |
| Dissipation factor | A<br>D-24/23<br>D-48/50 | 4.6.6 | Max. | | | | | | | | | | |
| Volume resistivity | C-96/35/90 | 4.6.7 | Min. megohm cm | | | | | | | | | | |
| Surface resistance | C-96/35/90 | 4.6.8 | Min. megohm | | | | | | | | | | |
| Arc resistance | D-48/50 | 4.6.9 | Min. seconds | | | | 6.5 | 6.5 | 6.5 | 6.5 | 6.5 | 6.5 | 6.5 |
| Tracking resistance | A | 4.6.10 | Min. minutes | | | | 5.5 | 5.5 | 5.5 | 5.5 | 5.5 | 5.5 | 5.5 |
| Impact strength[2]<br>Lengthwise:<br>Crosswise: | E-48/50 | 4.6.11 | Min. ft – lbs. per inch | | | | | | | | | | |
| Flexural strength<br>Lengthwise:<br>Crosswise: | A | 4.6.12 | Min. psi | 18,000<br>16,000 | 20,000<br>18,000 | 20,000<br>18,000 | 20,000<br>18,000 | 20,000<br>18,000 | 20,000<br>18,000 | 20,000<br>18,000 | 19,000<br>17,000 | 17,000<br>15,300 | 17,000<br>15,300 |

See footnotes a end of table.





Source: https://assist.dla.mil -- Downloaded: 202610-20T02:00Z
Check the source to verify that this is the current version before use.

TABLE V. *Performance requirements for sheets – Continued.*

| Requirement | Cond. | Test para. | Unit | Thickness (inches)[1] | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 0.031 | 0.062 | 0.094 | 0.125 | 0.188 | 0.250 | 0.500 | 0.750 | 1.000 | 1.001 – Max. |
| Thermal endurance Flexural strength: | E-200/150 | 4.6.13 | Min psi | | | | | | | | | | |
| Bonding strength[2] | D<br>D-48/50 | 4.6.15 | Min. pounds | | | | | | | 850<br>700 | 850<br>700 | 850<br>700 | 850<br>700 |
| Water absorption | $D_1$-24/23 | 4.6.17 | Max. percent | 4.20 | 2.70 | 2.30 | 2.00 | 1.90 | 1.80 | 1.50 | 1.25 | 1.00 | 1.00 |
| Silicone content | E-168/18 | 4.6.18 | Max. ppm | | | | | | | | | | |
| Flame resistance | | | | | | | | | | | | | |
| Ignition time: | A | 4.6.19 | Min. sec. | | | | | | | | | | |
| Burning time: | | | Max. sec. | | | | | | | | | | |

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]*Maximum thickness tested shall be 2.000 inches.*





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:00Z
Check the source to verify that this is the current version before use.

TABLE VI. *Performance requirements for rolled tubes.*

| Requirement | Cond. | Test para. | Unit | Wall thickness (inches) with inside diameter 0.125–8.000 inches | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 0.031–0.047 | 0.048–0.062 | 0.063–0.094 | 0.095–0.125 | 0.126–0.188 | 0.189–0.250 |
| Dielectric strength | A D-48/50 | 4.6.5 | Min. volts per mil | | | | | | |
| Compressive strength | A | 4.6.14 | Min. psi | | | | | | |
| Specific gravity | A | 4.6.16 | Min. | | | | | | |
| Water absorption | $D_1$-24/23 | 4.6.17 | Max. percent | | | | | | |

TABLE VI. *Performance requirements for rolled tubes – Continued.*

| Requirement | Cond. | Test para. | Unit | Wall thickness (inches) with inside diameter 0.125–8.000 inches | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | 0.251–0.375 | 0.376–0.500 | 0.501–0.750 | 0.751–1.000 | 1.001 Max. |
| Dielectric strength | A D-48/50 | 4.6.5 | Min. volts per mil | | | | | |
| Compressive strength | A | 4.6.14 | Min. psi | | | | | |
| Specific gravity | A | 4.6.16 | Min. | | | | | |
| Water absorption | $D_1$-24/23 | 4.6.17 | Max. percent | | | | | |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:00Z
Check the source to verify that this is the current version before use.

TABLE VII. *Performance requirements for rods.*

| Requirement | Cond. | Test para. | Unit | Diameter (inches) | | | |
|---|---|---|---|---|---|---|---|
| | | | | 0.250–0.499 | 0.500–0.999 | 1.000 | 1.001–2.000 |
| Flexural strength | A | 4.6.12 | Min. psi | | | | |
| Compressive strength | A | 4.6.14 | Min. psi | | | | |
| Specific gravity | A | 4.6.16 | Min. | | | | |
| Water absorption | $D_1$-24/23 | 4.6.17 | Max. percent | | | | |





Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:00Z
Check the source to verify that this is the current version before use.

Custodians:
Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
(Project 5970-1015-18)



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:00Z
Check the source to verify that this is the current version before use.